# Judging criteria → evidence in this entry

(For the author; not part of the Community post.)

| Criterion | What makes it strong | Where |
|---|---|---|
| **Originality / impact** | Physics-trajectory recalibration: GENCO's own per-layer AC mismatch explains 56 % of its error, turning a neural solver into a screen with a finite-sample guarantee. First certified (Benders) solution of the RSS bilevel; shows DC overstates the secure range by 43 % and that 45/48 DC limits are AC-insecure. | post §1, §3; paper Sec. III–IV |
| **Technical quality** | Every equation of the code written out and every claim proved (RSS-interval theorem, vertex lemma, Benders relaxation/certificate/termination, phase-2 exactness, nesting of AC corrections, split-conformal no-miss guarantee; paper appendix and `paper/supplement.pdf`), each also checked numerically in `tests/test_theory.py`; 20-split validation (observed miss 0.09 % ≤ 5 %); full exact-AC audit of 19,500 pool scenarios (1 miss); big-M baseline re-checked against true bilevel values; unit tests; every number generated by code. | `experiments/`, `tests/`, `results/numbers.json` |
| **Exploration of GENCO** | Trained with gridfm-graphkit on gridfm-datakit data of the RSS region; corrective-layer probe via forward hook; recalibration from the trajectory; unseen-topology test (56 % cleared); autograd adversary over the PV/load box (41/100 wins). | `genco_rss/genco.py`, E3, E4 |
| **Communication** | One-page TL;DR, six figures, one table, method diagram, explicit limitations (no speed claim: exact AC 0.29 ms vs GENCO 10.3 ms per scenario on CPU). | `docs/submission_post.md`, `paper/main.pdf` |
| **Open science** | One command reproduces everything; trained weights shipped; Gradio Space with live | `genco_ac_rssr_challenge.py`, `app.py`, `weights/` |

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|---|---|---|
| | Benders and live conformal screen; Apache-2.0. | |

Before posting: create the Space, upload this folder, then `SPACE_URL=… CODE_URL=… PAPER_URL=… IMG_BASE=https://huggingface.co/spaces/<user>/<space>/resolve/main/results/figures python experiments/e8_docs.py` and paste `docs/submission_post.md` into the challenge Community tab before 25 Oct 2026 (AoE).